

# Paper I - Basic Sciences in Urology (MCh Urology, Summer 2015)

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## I-1. Describe in details basic patho-physiology of urine production and transport

**Type:** LAQ (25 marks)

### Introduction

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- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled by peristalsis into a low-pressure bladder.

### A. Renal blood flow (RBF)

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- RBF is **20-25% of cardiac output (CO)**; renal plasma flow (RPF) ~600 mL/min.
- The medulla receives <10% of flow; this protects its osmotic gradient but makes it prone to hypoxia.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure **80-180 mmHg**:
  - **Myogenic**: afferent arteriolar smooth muscle contracts when stretched.
  - **Tubuloglomerular feedback (TGF)**: high NaCl at the macula densa releases adenosine, which constricts the afferent arteriole and reduces renin.
- Non-steroidal anti-inflammatory drugs (NSAIDs) remove protective prostaglandin vasodilatation in low-flow states.

### B. Glomerular filtration

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- GFR is **~125 mL/min (~180 L/day)**; filtration fraction (FF = GFR/RPF) is about **20%**.
- **Starling equation**:  $GFR = K_f \times [(P_{GC} - P_{BS}) - (\pi_{GC} - \pi_{BS})]$ . Glomerular capillary pressure is the main driving force; **obstruction** raises Bowman's space pressure and so lowers GFR.
- The filtration barrier restricts molecules by size (>~60-70 kDa) and negative charge, so albumin is excluded.

### C. Tubular handling - segment-wise

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| Segment                                 | Handling                                                                                                                                         | Clinical link    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Proximal convoluted tubule (PCT)</b> | ~ <b>65%</b> of Na <sup>+</sup> and water (iso-osmotic); ~100% of glucose via SGLT2 (sodium-glucose cotransporter 2); secretes drugs, creatinine | SGLT2 inhibitors |

| Segment                                       | Handling                                                                                                                                                                | Clinical link                                                  |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Thick ascending limb (TAL)</b>             | ~ <b>25%</b> of NaCl via NKCC2 ( $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter); water-impermeable " <b>diluting segment</b> " that builds the medullary gradient | Loop diuretics                                                 |
| <b>Distal convoluted tubule (DCT)</b>         | ~5–7% of $\text{Na}^+$ ; $\text{Ca}^{2+}$ reabsorption under parathyroid hormone (PTH)                                                                                  | Thiazides lower urinary calcium                                |
| <b>Collecting duct (CD) - principal cells</b> | $\text{Na}^+$ reabsorption/ $\text{K}^+$ secretion (aldosterone); water via aquaporin-2 (AQP2) under antidiuretic hormone (ADH)                                         | Amiloride, spironolactone                                      |
| Intercalated cells                            | Acid secretion                                                                                                                                                          | Distal renal tubular acidosis (RTA) → calcium phosphate stones |

## D. Concentration and dilution

- **Countercurrent multiplication:** the TAL pumps out NaCl without water; this small single effect (~200 mOsm) is multiplied along the loop, raising papillary interstitial osmolality to ~**1200 mOsm/kg**.
- ADH-driven **urea recycling** supplies about half of inner-medullary osmolality.
- The **vasa recta** (hairpin loops with slow flow) act as countercurrent exchangers, so they do not wash out the gradient.
- **ADH action:** released when osmolality rises (~1–2%) or volume falls; via the **V2 receptor** and cAMP (cyclic adenosine monophosphate) it inserts **AQP2** channels into the apical membrane, so water is reabsorbed.
- Nephrogenic diabetes insipidus: lithium, hypercalcaemia and **post-obstruction** (reduced AQP2).

## E. Hormonal control

- Angiotensin II (Ang II) and aldosterone retain  $\text{Na}^+$  and maintain GFR; ADH reabsorbs free water.
- Atrial natriuretic peptide (ANP) causes natriuresis; PTH causes phosphaturia.

## F. Upper urinary tract transport

- **Pacemaker: atypical smooth muscle cells (SMCs)** at the pelvicalyceal border; interstitial cells of Cajal-like cells act as secondary pacemakers.
- Peristalsis is **myogenic** (persists after denervation and transplantation); nerves only modulate it.
- **Electromechanical coupling:** L-type calcium entry creates the action potential, spreading via gap junctions; calcium-calmodulin activates myosin light-chain kinase → contraction.
- **Bolus propagation:** the ureter contracts behind each bolus with walls fully **coapted**; **2–6 waves/min**, ~20–80 cmH<sub>2</sub>O (resting ~0–5 cmH<sub>2</sub>O).
- **Effect of flow:** higher flow first increases frequency, then bolus size. At very high flow boluses **coalesce** into a fluid column and the ureter dilates **without obstruction** (relevant to diuretic renography).
- **Ureterovesical junction (UVJ):** an intramural tunnel with **length:diameter ≈ 5:1** (Paquin) prevents reflux. Sustained bladder pressure >~40 cmH<sub>2</sub>O (neurogenic bladder) stops efflux and causes secondary hydronephrosis.
- **Factors:** Gram-negative endotoxin inhibits peristalsis; **α1-blockers** (tamsulosin) relax the ureter for medical expulsive therapy.

## G. Bladder storage

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- High compliance, sympathetic detrusor relaxation and sphincter tone let the bladder hold ~400–500 mL at  $< \sim 15$  cmH<sub>2</sub>O.

## H. Pathophysiology of obstruction

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### Unilateral ureteral obstruction (UUO) - triphasic response:

|                    |                                    |                                   |
|--------------------|------------------------------------|-----------------------------------|
| Phase 1 (0–1.5 h): | RBF rises, ureteral pressure rises | (afferent dilatation)             |
| Phase 2 (1.5–5 h): | RBF falls, pressure still rises    | (efferent constriction)           |
| Phase 3 (>5 h):    | RBF falls, pressure falls          | (afferent constriction by Ang II) |

- GFR falls because RBF drops and tubular back-pressure opposes filtration.
- In **bilateral obstruction or a solitary kidney**, pressure stays high longer and natriuretic substances (ANP, urea) accumulate, causing **post-obstructive diuresis** after relief.
- Chronic obstruction causes fibrosis via Ang II and transforming growth factor- $\beta$ 1; recovery depends on duration and degree.

**Diagram to draw:** Nephron with segmental transporters and % Na<sup>+</sup> reabsorbed; triphasic UUO graph.

### Recent advances

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- SGLT2 inhibitors restore tubuloglomerular feedback and protect the kidney.
- Tolvaptan (a V2 antagonist) is used in autosomal dominant polycystic kidney disease.

#### Key points to remember:

- RBF 20–25% of CO; GFR 125 mL/min; FF 20%; autoregulation 80–180 mmHg.
  - PCT 65%, TAL 25%, DCT 5–7%; ADH: V2  $\rightarrow$  cAMP  $\rightarrow$  AQP2.
  - Pacemaker: atypical SMCs; peristalsis myogenic, 2–6/min.
  - UUO triphasic: RBF/pressure  $\uparrow/\uparrow$ ,  $\downarrow/\uparrow$ ,  $\downarrow/\downarrow$ .
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## I-2. Describe female pelvic anatomy in relation to prolapse and SUI

Type: LAQ (25 marks)

### Introduction

- Support depends on **levator ani, endopelvic fascia/ligaments and bony pelvis** acting together. Failure causes **pelvic organ prolapse (POP)** and **stress urinary incontinence (SUI)** – effort leakage when urethral support or closure fails.

### A. Bony pelvis

- Standing, the organs rest on the **levator plate**, not over the hiatus. Surgical landmarks: **ischial spine, sacrospinous** and **Cooper's ligaments**, sacral promontory.

### B. Pelvic floor muscles (pelvic diaphragm)

| Muscle                       | Attachment                                   | Function                                    |
|------------------------------|----------------------------------------------|---------------------------------------------|
| Pubococcygeus (pubovisceral) | Pubis to vagina, perineal body, anus         | Closes the hiatus, lifts urethra and vagina |
| Puborectalis                 | Sling behind the anorectal junction          | Maintains anorectal angle                   |
| Iliococcygeus                | Arcus tendineus levator ani (ATLA) to coccyx | Forms the horizontal <b>levator plate</b>   |

- Levator avulsion** at vaginal delivery is a strong risk factor for POP and recurrence.

### C. Endopelvic fascia and ligaments

- Cardinal and uterosacral ligaments** suspend the apex.
- Pubocervical fascia** supports the anterior wall and attaches laterally to the **arcus tendineus fasciae pelvis (ATFP, "white line")**, running from pubis to ischial spine.
- Rectovaginal fascia** supports the posterior wall; **pubourethral ligaments** anchor the mid-urethra to the pubis.

### D. DeLancey's three levels of support (1992)

| Level                  | Region              | Structures                                            | Defect causes                            |
|------------------------|---------------------|-------------------------------------------------------|------------------------------------------|
| <b>I - suspension</b>  | Cervix/upper vagina | Cardinal-uterosacral complex                          | Uterine/vault prolapse, enterocele       |
| <b>II - attachment</b> | Mid vagina          | ATFP (anterior), ATLA/rectovaginal fascia (posterior) | Paravaginal cystocele, rectocele         |
| <b>III - fusion</b>    | Distal 2–3 cm       | Perineal membrane, perineal body                      | Urethral hypermobility, perineal descent |

### E. Urethral support and continence mechanisms

- Hammock hypothesis (DeLancey 1994)**: the urethra lies on the anterior vaginal wall and endopelvic fascia. A rise in abdominal pressure compresses it against this firm layer; loss of the hammock causes SUI.
- Integral theory (Petros & Ulmsten 1990)**: SUI and urgency both arise from laxity of the vagina and its ligaments. The **mid-urethral pubourethral ligament** is key, which is the basis of the **tension-free vaginal tape (TVT, 1996)**.
- Intrinsic sphincter deficiency (ISD)** (Valsalva leak point pressure <60 cmH<sub>2</sub>O or maximal urethral closure pressure <20 cmH<sub>2</sub>O) does poorly with colposuspension; slings or

bulking are preferred.

- **Intrinsic closure:** mucosal coaptation (oestrogen-dependent), smooth muscle and rhabdosphincter.
- SUI is a continuum of **hypermobility with or without ISD**.

## F. Compartments and defects

| Compartment | Defect                             | Anatomical basis                                                  |
|-------------|------------------------------------|-------------------------------------------------------------------|
| Anterior    | <b>Central cystocele</b>           | Midline thinning of pubocervical fascia; <b>rugae lost</b>        |
|             | <b>Paravaginal cystocele</b>       | Detachment from ATRP; <b>rugae preserved</b> , lateral sulci lost |
| Apical      | Uterine/vault prolapse, enterocele | Level I failure                                                   |
| Posterior   | Rectocele, perineal descent        | Rectovaginal fascia or perineal body defect                       |

- Apical support underpins the anterior wall, so an anterior repair fails if the apex is not corrected.

## G. Innervation

- The **pudendal nerve (S2-4)** supplies the sphincters and is stretched during delivery.
- The **levator ani nerve (S3-5)** runs on the pelvic surface of the muscle, separate from the pudendal nerve.

## H. Pathophysiology - risk factors (Bump & Norton)

- **Predisposing:** genetics and connective tissue disorders (Ehlers-Danlos, Marfan).
- **Inciting:** **vaginal childbirth** (strongest factor) and hysterectomy.
- **Promoting:** obesity, chronic cough, constipation.
- **Decompensating:** ageing and **menopause**.

## I. POP-Q staging (Pelvic Organ Prolapse Quantification; International Continence Society, Bump 1996)

- **Hymen = 0**; points above are negative, below positive (cm). Nine measurements, including total vaginal length (tvL).

| Stage | Definition                           |
|-------|--------------------------------------|
| 0     | No prolapse                          |
| I     | Most distal point >1 cm above hymen  |
| II    | Within $\pm 1$ cm of hymen           |
| III   | >1 cm below hymen but < (tvL - 2) cm |
| IV    | Complete eversion ( $\geq$ tvL - 2)  |

## Clinical assessment

- Examine standing with Valsalva, Sims speculum per compartment; **cough stress test**, also with prolapse reduced (occult SUI); **Q-tip** change  $>30^\circ$  = hypermobility.

## J. How anatomy guides surgery

| Defect              | Operation                                                          | Anatomical principle           |
|---------------------|--------------------------------------------------------------------|--------------------------------|
| SUI - hypermobility | <b>Mid-urethral sling (MUS)</b> (retropubic TVT or transobturator) | Recreates pubourethral support |

| Defect                 | Operation                                                                                              | Anatomical principle                            |
|------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| SUI                    | <b>Burch colposuspension</b>                                                                           | Periurethral fascia to <b>Cooper's ligament</b> |
| SUI – ISD              | Autologous sling, bulking, artificial urinary sphincter                                                | Compression/coaptation                          |
| Paravaginal cystocele  | Paravaginal repair                                                                                     | Reattaches to ATFP                              |
| Central cystocele      | Anterior colporrhaphy                                                                                  | Midline plication                               |
| Apical                 | <b>Sacrocolpopexy</b> (gold-standard durability)                                                       | Mesh to the sacral promontory                   |
| Apical (vaginal route) | <b>Sacrospinous fixation</b> ~1.5–2 cm medial to the ischial spine (avoids pudendal vessels and nerve) | Level I restoration                             |

## K. Mesh regulation

- In **April 2019 the US Food and Drug Administration (FDA)** stopped sale of **transvaginal mesh for anterior POP repair**. This did **not** apply to MUS or sacrocolpopexy mesh.

**Diagram to draw:** DeLancey levels I–III; hammock cross-section with a pressure arrow.

## Recent advances

- 3D/4D translabial ultrasound detects levator avulsion, predicting recurrence.
- Single-incision mini-slings give short-term results comparable to standard MUS (SIMS randomised trial, 2022).

### Key points to remember:

- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Hammock (DeLancey 1994) vs Integral theory (Petros & Ulmsten 1990 → TVT 1996).
- Central cystocele rugae lost; paravaginal preserved. POP-Q Stage II =  $\pm 1$  cm of hymen.
- FDA 2019: anterior transvaginal mesh withdrawn; MUS unaffected.

## I-3a. Newer generation CT scan

**Type:** SAQ (10 marks)

### Introduction

- Computed tomography (CT) is the reference imaging for stones, renal masses, haematuria and trauma.

### Evolution

| Generation                                              | Feature                                                           |
|---------------------------------------------------------|-------------------------------------------------------------------|
| Conventional (Hounsfield, 1971-72)                      | Step-and-shoot axial images                                       |
| <b>Helical/spiral CT</b> (~1989)                        | Continuous table movement gives volumetric data                   |
| <b>Multidetector CT (MDCT)</b> (4-slice 1998 → 320-row) | Sub-millimetre slices, 3D reconstruction in one breath-hold       |
| <b>Dual-source CT</b> (2005)                            | Two tube-detector pairs: speed and dual energy                    |
| <b>Photon-counting CT (PCCT)</b>                        | First US Food and Drug Administration (FDA) clearance <b>2021</b> |

### Dual-energy CT (DECT)

- Each material attenuates the two energy levels in a characteristic ratio, allowing material decomposition: **uric acid vs non-uric-acid stones** (oral chemolysis candidates).
- **Virtual non-contrast** images replace the unenhanced phase; **iodine maps** separate cysts from solid masses.

### Low-dose CT KUB (kidney-ureter-bladder)

- Standard CT gives ~**8-10 mSv**; low-dose CT KUB **<3-4 mSv**; ultra-low-dose **<1 mSv**.
- Stone sensitivity/specificity remain ~95%.
- The European Association of Urology (EAU) recommends **low-dose non-contrast CT** for acute flank pain (after ultrasound) and follow-up.

### CT urography (CTU)

- For visible haematuria/upper tract cancer. **Triple-phase** (unenhanced, nephrographic, excretory) gives ~15-25 mSv.
- **Split-bolus** CTU gives contrast in two aliquots and combines phases in one acquisition, lowering dose.

### Dose reduction technology

- Tube current modulation and **iterative/deep-learning reconstruction** cut dose while keeping image quality.

### Photon-counting CT

- The detector counts individual photons and sorts their energy: **no electronic noise**, higher resolution, lower dose.

### Other advances

- CT angiography and 3D models for donor and partial nephrectomy planning.

## Uses in urology

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- Skin-to-stone distance >10 cm predicts shock wave lithotripsy failure; also Bosniak 2019, staging, trauma.

**Diagram to draw:** Timeline: axial → helical → MDCT → dual-source → PCCT.

## Recent advances

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- PCCT in clinical use since 2021; deep-learning reconstruction.

### Key points to remember:

- Helical ~1989; MDCT 1998; dual-source 2005; PCCT FDA-cleared 2021.
  - DECT: uric acid stones, virtual non-contrast, iodine maps.
  - Low-dose KUB <3–4 mSv (EAU-preferred).
  - Split-bolus CTU reduces phases and dose.
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## I-3b. History of Laparoscopy

**Type:** SAQ (10 marks)

### Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum. It evolved from optics to diagnostic, therapeutic and finally robotic surgery.

### Timeline

| Year        | Pioneer                       | Milestone                                                                                     |
|-------------|-------------------------------|-----------------------------------------------------------------------------------------------|
| <b>1806</b> | <b>Bozzini</b>                | <b>Lichtleiter</b> (candle-lit light conductor) – birth of endoscopy                          |
| 1877–79     | Nitze                         | Cystoscope with distal illumination and lenses                                                |
| <b>1901</b> | <b>Kelling</b>                | Viewed the air-insufflated abdomen of a <b>dog</b>                                            |
| <b>1910</b> | <b>Jacobaeus</b>              | First human series; coined "laparoscopy"                                                      |
| 1924        | Zollikofer                    | CO <sub>2</sub> for insufflation                                                              |
| <b>1938</b> | <b>Veress</b>                 | Spring-loaded needle                                                                          |
| 1950s       | <b>Hopkins</b>                | <b>Rod-lens system</b> giving bright, clear images                                            |
| 1970s–80s   | <b>Semm</b>                   | Automatic CO <sub>2</sub> insufflator; first laparoscopic appendectomy (1980)                 |
| 1970s       | Hasson                        | Open access technique                                                                         |
| <b>1976</b> | <b>Cortesi</b>                | Localised <b>impalpable testis</b> – first urological use                                     |
| 1985        | Mühe                          | First laparoscopic cholecystectomy                                                            |
| <b>1987</b> | <b>Mouret</b>                 | First video-laparoscopic cholecystectomy                                                      |
| <b>1990</b> | <b>Clayman</b>                | <b>First laparoscopic nephrectomy</b>                                                         |
| 1992        | Gagner                        | Laparoscopic adrenalectomy                                                                    |
| <b>1992</b> | <b>Schuessler</b>             | <b>First laparoscopic radical prostatectomy</b> (reported 1997, 9 patients)                   |
| 1993        | Schuessler; Kavoussi & Peters | Laparoscopic dismembered pyeloplasty                                                          |
| <b>1995</b> | <b>Ratner</b>                 | Laparoscopic live-donor nephrectomy                                                           |
| 1998–99     | Guillonneau & Vallancien      | Standardised and revived laparoscopic prostatectomy                                           |
| <b>2000</b> | <b>da Vinci</b>               | US Food and Drug Administration (FDA) clearance                                               |
| <b>2000</b> | <b>Binder</b>                 | <b>First robot-assisted radical prostatectomy (RARP)</b>                                      |
| Early 2000s | <b>Menon</b>                  | Vattikuti Institute Prostatectomy (VIP) standardised RARP                                     |
| 2004        | Kalloo                        | Transgastric peritoneoscopy → NOTES (natural orifice transluminal endoscopic surgery) concept |
| 2007        | Raman, Rane                   | Single-port nephrectomy                                                                       |
| 2018        | da Vinci SP                   | Single-port robot cleared for urology                                                         |

### Phases of evolution

Optics → Diagnostic → Therapeutic (Semm) → Video laparoscopy  
 → Urological surgery → Robotic → Single-site / telesurgery

## Indian contribution

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- Rapid uptake of urological laparoscopy and robotics; an indigenous robotic system (**SSI Mantra**) was developed in India, with telesurgery demonstrations.

## Recent advances

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- New multi-port robots (Versius, Hugo RAS, Hinotori), long-distance telesurgery and haptic feedback (da Vinci 5).

**Diagram to draw:** Timeline arrow 1806 → 1901 → 1910 → 1938 → 1987 → 1990 → 2000.

### Key points to remember:

- Bozzini 1806 Lichtleiter; Kelling 1901 (dog); Jacobaeus 1910 (human).
  - Veress 1938; Hopkins rod-lens; Semm – father of modern operative laparoscopy.
  - Urology: Cortesi 1976; Clayman 1990 nephrectomy; Ratner 1995.
  - da Vinci FDA 2000; Binder 2000 first RARP; Menon – VIP.
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## I-3c. Renin-angiotensin system

**Type:** SAQ (10 marks)

### Definition

- Hormonal cascade regulating **blood pressure, extracellular volume, Na<sup>+</sup>/K<sup>+</sup> balance and glomerular filtration rate (GFR)**.

### Cascade

Angiotensinogen (liver)  
↓ renin (juxtaglomerular cells) – rate-limiting step  
Angiotensin I  
↓ angiotensin-converting enzyme (ACE), mainly in lung  
Angiotensin II → AT1 and AT2 receptors  
↓ ACE2  
Angiotensin (1-7) → Mas receptor (vasodilator, anti-fibrotic)

### Juxtaglomerular (JG) apparatus

- **JG cells** in the afferent arteriole secrete renin and act as baroreceptors.
- The **macula densa** senses tubular NaCl.

### Stimuli for renin release

- **Stimulated by** low perfusion pressure, low macula densa NaCl, **sympathetic β1 activity** and hypovolaemia.
- **Inhibited by** angiotensin II (negative feedback), atrial natriuretic peptide and β-blockers.

### Receptors and actions

- **AT1 (angiotensin II type 1) receptor:** vasoconstriction, **aldosterone** release (Na<sup>+</sup> retention, K<sup>+</sup> loss), thirst and **fibrosis**.
- **AT2 receptor:** vasodilatation; important in fetal urinary tract development.
- The **ACE2-angiotensin (1-7)** arm is protective and counter-regulatory.

### Intrarenal effects on GFR

- Angiotensin II constricts the **efferent more than the afferent** arteriole, maintaining GFR when perfusion is low.
- Therefore **ACE inhibitors and angiotensin receptor blockers cause an acute fall in GFR** in bilateral renal artery stenosis or stenosis of a solitary kidney.

### Urological relevance

- **Renovascular hypertension** (Goldblatt 1934): atherosclerosis or fibromuscular dysplasia. A **renal vein renin ratio >1.5** predicts response to revascularisation or nephrectomy.
- **Page kidney:** perirenal compression (e.g. subcapsular haematoma) causes renin-mediated hypertension.
- **Reninoma (JG cell tumour):** young adults with severe hypertension and **hypokalaemia; partial nephrectomy is curative**.
- **Obstruction:** angiotensin II mediates late afferent vasoconstriction and fibrosis.
- **Drugs:** ACE inhibitors, angiotensin receptor blockers and aliskiren are contraindicated in pregnancy.

**Diagram to draw:** JG apparatus and the cascade flowchart.

## Recent advances

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- Renal denervation (US Food and Drug Administration approval 2023) for resistant hypertension.

### Key points to remember:

- Renin (JG cells) is rate-limiting; released by low pressure, low NaCl,  $\beta 1$ .
  - Efferent constriction maintains GFR → ACE inhibitor risk in bilateral stenosis.
  - AT1 = pressor/fibrotic; AT2 and ACE2-Ang(1-7) = protective.
  - Renin ratio >1.5 lateralises; reninoma → partial nephrectomy.
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## I-3d. Patho-physiology of Brain stem death

**Type:** SAQ (10 marks)

### Definition

- **Irreversible loss of the capacity for consciousness and of the capacity to breathe** from permanent brainstem failure; legally death. Main source of donation after brainstem death (DBD) kidneys.

### Pathophysiology

Brain injury → oedema → raised intracranial pressure (ICP)  
→ falling cerebral perfusion → herniation ("coning")  
→ Pons: Cushing reflex (hypertension, bradycardia)  
→ Medulla: catecholamine storm (hypertension, arrhythmia)  
→ Loss of sympathetic tone → hypotension

- **Diabetes insipidus (DI)** from loss of antidiuretic hormone causes polyuria and hypernatraemia.
- Loss of hypothalamic control causes **hypothermia**; necrotic brain releases thromboplastin, causing **disseminated intravascular coagulation (DIC)**.
- A systemic inflammatory response increases delayed graft function and rejection in DBD kidneys.

### Diagnosis

- **Preconditions:** apnoeic ventilated coma with a known irreversible cause. **Exclude** sedatives, neuromuscular blockers, **hypothermia (<35 °C)** and metabolic disturbance.
- **All brainstem reflexes absent:** pupillary, corneal, caloric (~50 mL ice-cold water), doll's eye, cranial motor response to pain, gag and cough.
- **Apnoea test:** after pre-oxygenation, the ventilator is disconnected. No breathing effort when **arterial CO<sub>2</sub> (PaCO<sub>2</sub>) reaches ≥60 mmHg** confirms apnoea.

### Indian legal framework

- Recognised by the **Transplantation of Human Organs Act (THOA) 1994**, amended 2011, with Rules in 2014.
- Certified by a **4-member board** (hospital in-charge, independent doctor, neurologist/neurosurgeon, treating doctor); no transplant team members.
- **Two sets of tests at least 6 hours apart** are required.

### Donor management

- **Rule of 100s:** systolic blood pressure >100 mmHg, urine >100 mL/h, PaO<sub>2</sub> (arterial O<sub>2</sub> tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** is the preferred vasopressor because it also treats DI.

**Diagram to draw:** Herniation sequence leading to storm, DI, hypothermia and DIC.

### Recent advances

- Hypothermic machine perfusion of DBD kidneys reduces delayed graft function.

**Key points to remember:**

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
  - Exclude drugs, hypothermia ( $<35^{\circ}\text{C}$ ), metabolic causes.
  - Apnoea test: no breathing at  $\text{PaCO}_2 \geq 60$  mmHg.
  - THOA 1994: 4-member board, two tests  $\geq 6$  h apart; Rule of 100s.
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## I-3e. Male reproductive immunology

**Type:** SAQ (10 marks)

### Introduction

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- Sperm appear at puberty, after immune tolerance is established, so their antigens are "foreign"; the testis is therefore **immune-privileged**.

### Mechanisms of immune privilege

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- **Blood-testis barrier (BTB):** formed at **puberty** by **tight junctions between Sertoli cells**. It separates a **basal compartment** (spermatogonia, exposed to blood) from an **adluminal compartment** (later germ cells, sequestered).
- The barrier is weakest at the rete testis and epididymis, common sites of antigen leak.
- **Active immunoregulation:** anti-inflammatory macrophages, regulatory T cells, Sertoli-cell TGF- $\beta$  (transforming growth factor- $\beta$ ) and immunosuppressive **androgens**. The role of Fas ligand is debated.

### Antisperm antibodies (ASA)

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#### Causes

- **Vasectomy** (most common), obstruction, testicular trauma or torsion, biopsy and infection.

#### Effects

- Sperm agglutination, poor motility, failure to penetrate **cervical mucus** and failure of zona binding. Head-directed antibodies matter most.

#### Testing

- **Mixed antiglobulin reaction (MAR)** test on fresh semen or **immunobead test (IBT)** on washed sperm.
- By **WHO (World Health Organization) 2010**,  **$\geq 50\%$  of motile sperm with bound particles** is clinically significant.
- Indications: isolated asthenozoospermia, agglutination, unexplained infertility.

#### Treatment

- Corticosteroids give inconsistent benefit and risk **avascular necrosis**, so they are not recommended.
- Sperm washing with intrauterine insemination bypasses cervical mucus, with modest benefit.
- **ICSI (intracytoplasmic sperm injection) is preferred**, as fertilisation is largely unaffected by ASA.

### Inflammatory states

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- **Leukocytospermia** is  **$>1 \times 10^6$  peroxidase-positive leukocytes/mL**. Reactive oxygen species damage sperm DNA; treat infection if found.
- **Orchitis:** viral (post-pubertal mumps  $\rightarrow$  atrophy), bacterial (*Chlamydia*/gonorrhoea  $<35$  y; *E. coli* older) or granulomatous.
- **Autoimmune orchitis** may be primary (ASA with azoospermia) or secondary to systemic vasculitis.

**Diagram to draw:** Seminiferous tubule showing Sertoli tight junctions dividing basal and adluminal compartments.

### Recent advances

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- Sperm DNA fragmentation testing in men with ASA or leukocytospermia.

**Key points to remember:**

- BTB = Sertoli-Sertoli tight junctions; basal vs adluminal compartments; formed at puberty.
  - Privilege is also active (regulatory T cells, TGF- $\beta$ , androgens).
  - ASA: vasectomy commonest;  $\geq 50\%$  bound = significant; ICSI preferred.
  - Leukocytospermia  $>1$  million/mL.
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## I-3f. Principles of antimicrobial therapy in UTI

Type: SAQ (10 marks)

### Classification (EAU 2024)

- European Association of Urology (EAU) classes of urinary tract infection (UTI):
  - Uncomplicated:** non-pregnant women without abnormality or comorbidity.
  - Complicated:** men, pregnancy, abnormality, catheter, immunosuppression.
  - Recurrent:**  $\geq 2$  infections in 6 months or  $\geq 3$  in 12 months.

### Principles

- Confirm the diagnosis and send a culture**, except in simple cystitis in women.
- Pharmacodynamics:**  $\beta$ -lactams are **time-dependent** (need frequent dosing); **aminoglycosides and fluoroquinolones (FQs)** are **concentration-dependent** (high once-daily dose).
- Nitrofurantoin and fosfomycin reach **high urine but low tissue levels**, so they treat **cystitis only**, never pyelonephritis or urosepsis.
- Use an empirical drug only if local resistance is **<20%** for cystitis, and FQs for pyelonephritis only if resistance is **<10%**.
- Avoid FQs/cephalosporins in simple cystitis (collateral damage: resistance, *C. difficile*).
- Shortest effective course, de-escalate to culture, and **source control** (drain an obstructed infected kidney).

### Empirical choices

| Scenario                      | First-line                                                   | Duration                       |
|-------------------------------|--------------------------------------------------------------|--------------------------------|
| <b>Cystitis (women)</b>       | <b>Fosfomycin 3 g, nitrofurantoin or pivmecillinam</b>       | Single dose; <b>5 d; 3-5 d</b> |
| Alternative (resistance <20%) | Trimethoprim-sulfamethoxazole                                | 3 d                            |
| Cystitis in men               | Per culture                                                  | <b>7 d</b>                     |
| <b>Pyelonephritis (oral)</b>  | Ciprofloxacin, or levofloxacin 750 mg                        | 7 d; <b>5 d</b>                |
| <b>Complicated UTI</b>        | Aminoglycoside combination or 3rd-generation cephalosporin   | <b>7-14 d</b>                  |
| Catheter-associated           | Change catheter if in >2 weeks                               | <b>7 d</b>                     |
| Urosepsis                     | Broad-spectrum antibiotic <b>within 1 h</b> + source control | By response                    |

### Asymptomatic bacteriuria (ASB)

- Treat **only in pregnancy and before mucosa-breaching procedures**; not in elderly, diabetic or catheterised patients.

### Prophylaxis

- Recurrent UTI: vaginal oestrogen, **methenamine hippurate** (ALTAR 2022 – non-inferior to antibiotics), or low-dose/post-coital antibiotics.
- Peri-operative prophylaxis is a single dose or  $\leq 24$  hours.

### Recent advances

- Pivmecillinam (2024) and gepotidacin (2025) approved by the US Food and Drug Administration for uncomplicated UTI.

**Diagram to draw:** Concentration-time curve showing peak and time above the minimum inhibitory concentration.

**Key points to remember:**

- Recurrent =  $\geq 2$  in 6 months or  $\geq 3$  in 12 months.
  - Cystitis: fosfomicin 3 g single, nitrofurantoin 5 d, pivmecillinam 3–5 d.
  - Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d.
  - ASB: treat only in pregnancy/before mucosal surgery.
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## Rapid Revision Sheet - Paper I

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

### **I-1. Describe in details basic patho-physiology of urine production and transport (LAQ)**

- RBF 20–25% of CO; GFR 125 mL/min; FF 20%; autoregulation 80–180 mmHg.
- PCT 65%, TAL 25%, DCT 5–7%; ADH: V2 → cAMP → AQP2.
- Pacemaker: atypical SMCs; peristalsis myogenic, 2–6/min.
- UUU triphasic: RBF/pressure  $\uparrow/\uparrow$ ,  $\downarrow/\uparrow$ ,  $\downarrow/\downarrow$ .

### **I-2. Describe female pelvic anatomy in relation to prolapse and SUI (LAQ)**

- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Hammock (DeLancey 1994) vs Integral theory (Petros & Ulmsten 1990 → TVT 1996).
- Central cystocele rugae lost; paravaginal preserved. POP-Q Stage II =  $\pm 1$  cm of hymen.
- FDA 2019: anterior transvaginal mesh withdrawn; MUS unaffected.

### **I-3a. Newer generation CT scan (SAQ)**

- Helical ~1989; MDCT 1998; dual-source 2005; PCCT FDA-cleared 2021.
- DECT: uric acid stones, virtual non-contrast, iodine maps.
- Low-dose KUB  $<3\text{--}4$  mSv (EAU-preferred).
- Split-bolus CTU reduces phases and dose.

### **I-3b. History of Laparoscopy (SAQ)**

- Bozzini 1806 Lichtleiter; Kelling 1901 (dog); Jacobaeus 1910 (human).
- Veress 1938; Hopkins rod-lens; Semm – father of modern operative laparoscopy.
- Urology: Cortesi 1976; Clayman 1990 nephrectomy; Ratner 1995.
- da Vinci FDA 2000; Binder 2000 first RARP; Menon – VIP.

### **I-3c. Renin-angiotensin system (SAQ)**

- Renin (JG cells) is rate-limiting; released by low pressure, low NaCl,  $\beta_1$ .
- Efferent constriction maintains GFR → ACE inhibitor risk in bilateral stenosis.
- AT1 = pressor/fibrotic; AT2 and ACE2-Ang(1-7) = protective.
- Renin ratio  $>1.5$  lateralises; reninoma → partial nephrectomy.

### **I-3d. Patho-physiology of Brain stem death (SAQ)**

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Exclude drugs, hypothermia ( $<35^\circ\text{C}$ ), metabolic causes.
- Apnoea test: no breathing at  $\text{PaCO}_2 \geq 60$  mmHg.
- THOA 1994: 4-member board, two tests  $\geq 6$  h apart; Rule of 100s.

### **I-3e. Male reproductive immunology (SAQ)**

- BTB = Sertoli-Sertoli tight junctions; basal vs adluminal compartments; formed at puberty.
- Privilege is also active (regulatory T cells, TGF- $\beta$ , androgens).
- ASA: vasectomy commonest;  $\geq 50\%$  bound = significant; ICSI preferred.
- Leukocytospermia  $>1$  million/mL.

### **I-3f. Principles of antimicrobial therapy in UTI (SAQ)**

- Recurrent =  $\geq 2$  in 6 months or  $\geq 3$  in 12 months.
- Cystitis: fosfomycin 3 g single, nitrofurantoin 5 d, pivmecillinam 3–5 d.
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d.
- ASB: treat only in pregnancy/before mucosal surgery.